

Analysis of Circuits Containing Ideal Transformers

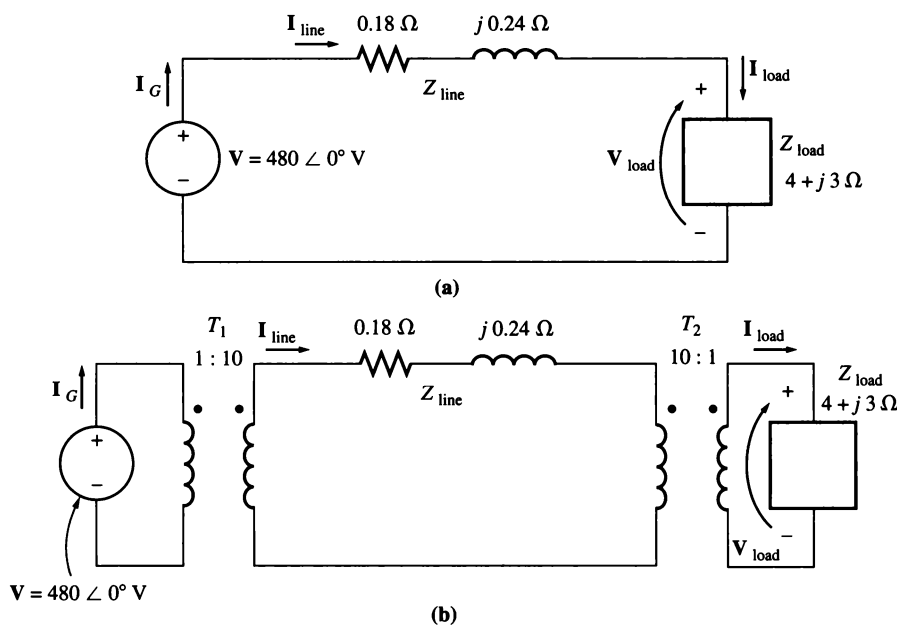
If a circuit contains an ideal transformer, then the easiest way to analyze the circuit for its voltages and currents is to replace the portion of the circuit on one side of the transformer by an equivalent circuit with the same terminal characteristics. After the equivalent circuit has been substituted for one side, then the new circuit (without a transformer present) can be solved for its voltages and currents. In the portion of the circuit that was not replaced, the solutions obtained will be the correct values of voltage and current for the original circuit. Then the turns ratio of the transformer can be used to determine the voltages and currents on the other side of the transformer. The process of replacing one side of a transformer with its equivalent at the other side's voltage level is known as *referring* the first side of the transformer to the second side.

How is the equivalent circuit formed? Its shape is exactly the same as the shape of the original circuit. The values of voltages on the side being replaced are scaled by Equation (2-4), and the values of the impedances are scaled by Equation (2-15). The polarities of voltage sources in the equivalent circuit will be reversed from their direction in the original circuit if the dots on one side of the transformer windings are reversed compared to the dots on the other side of the transformer windings.

The solution for circuits containing ideal transformers is illustrated in the following example.

Example 2-1. A single-phase power system consists of a 480-V 60-Hz generator supplying a load $Z_{\text{load}} = 4 + j3 \, \Omega$ through a transmission line of impedance $Z_{\text{line}} = 0.18 + j0.24 \, \Omega$. Answer the following questions about this system.

- (a) If the power system is exactly as described above (and shown in Figure 2-6a), what will the voltage at the load be? What will the transmission line losses be?

**FIGURE 2-6**

The power system of Example 2-1 (a) without and (b) with transformers at the ends of the transmission line.

- (b) Suppose a 1:10 step-up transformer is placed at the generator end of the transmission line and a 10:1 step-down transformer is placed at the load end of the line (as shown in Figure 2-6b). What will the load voltage be now? What will the transmission line losses be now?

Solution

- (a) Figure 2-6a shows the power system without transformers. Here $I_G = I_{\text{line}} = I_{\text{load}}$. The line current in this system is given by

$$\begin{aligned}
 I_{\text{line}} &= \frac{V}{Z_{\text{line}} + Z_{\text{load}}} \\
 &= \frac{480 \angle 0^\circ \text{ V}}{(0.18 \Omega + j0.24 \Omega) + (4 \Omega + j3 \Omega)} \\
 &= \frac{480 \angle 0^\circ}{4.18 + j3.24} = \frac{480 \angle 0^\circ}{5.29 \angle 37.8^\circ} \\
 &= 90.8 \angle -37.8^\circ \text{ A}
 \end{aligned}$$

Therefore the load voltage is

$$\begin{aligned}
 V_{\text{load}} &= I_{\text{line}} Z_{\text{load}} \\
 &= (90.8 \angle -37.8^\circ \text{ A})(4 \Omega + j3 \Omega) \\
 &= (90.8 \angle -37.8^\circ \text{ A})(5 \angle 36.9^\circ \Omega) \\
 &= 454 \angle -0.9^\circ \text{ V}
 \end{aligned}$$

and the line losses are

$$\begin{aligned} P_{\text{loss}} &= (I_{\text{line}})^2 R_{\text{line}} \\ &= (90.8 \text{ A})^2 (0.18 \Omega) = 1484 \text{ W} \end{aligned}$$

(b) Figure 2–6b shows the power system with the transformers. To analyze this system, it is necessary to convert it to a common voltage level. This is done in two steps:

1. Eliminate transformer T_2 by referring the load over to the transmission line's voltage level.
2. Eliminate transformer T_1 by referring the transmission line's elements and the equivalent load at the transmission line's voltage over to the source side.

The value of the load's impedance when reflected to the transmission system's voltage is

$$\begin{aligned} Z'_{\text{load}} &= a^2 Z_{\text{load}} \\ &= \left(\frac{10}{1}\right)^2 (4 \Omega + j3 \Omega) \\ &= 400 \Omega + j300 \Omega \end{aligned}$$

The total impedance at the transmission line level is now

$$\begin{aligned} Z_{\text{eq}} &= Z_{\text{line}} + Z'_{\text{load}} \\ &= 400.18 + j300.24 \Omega = 500.3 \angle 36.88^\circ \Omega \end{aligned}$$

This equivalent circuit is shown in Figure 2–7a. The total impedance at the transmission line level ($Z_{\text{line}} + Z'_{\text{load}}$) is now reflected across T_1 to the source's voltage level:

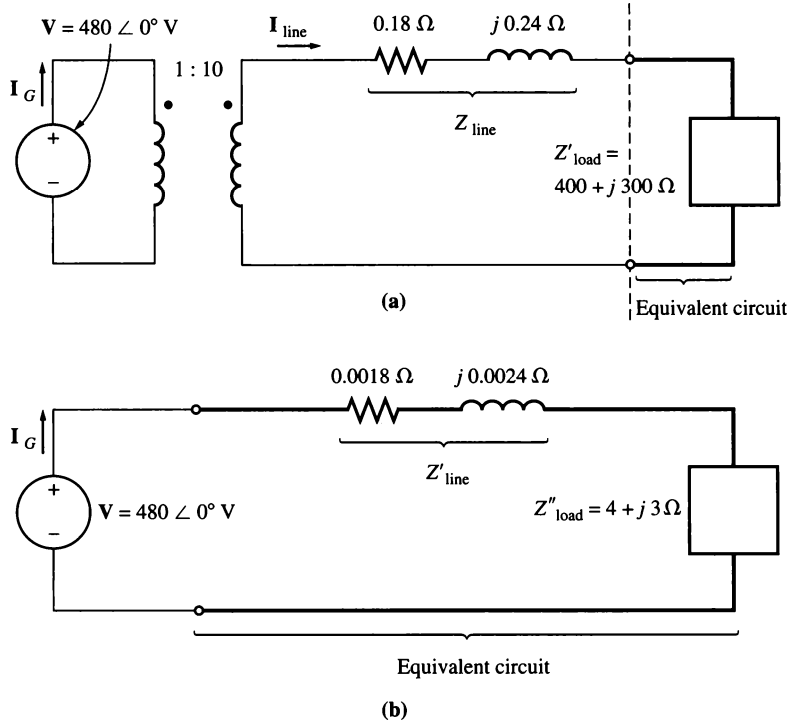
$$\begin{aligned} Z'_{\text{eq}} &= a^2 Z_{\text{eq}} \\ &= a^2 (Z_{\text{line}} + Z'_{\text{load}}) \\ &= \left(\frac{1}{10}\right)^2 (0.18 \Omega + j0.24 \Omega + 400 \Omega + j300 \Omega) \\ &= (0.0018 \Omega + j0.0024 \Omega + 4 \Omega + j3 \Omega) \\ &= 5.003 \angle 36.88^\circ \Omega \end{aligned}$$

Notice that $Z'_{\text{load}} = 4 + j3 \Omega$ and $Z'_{\text{line}} = 0.0018 + j0.0024 \Omega$. The resulting equivalent circuit is shown in Figure 2–7b. The generator's current is

$$I_G = \frac{480 \angle 0^\circ \text{ V}}{5.003 \angle 36.88^\circ \Omega} = 95.94 \angle -36.88^\circ \text{ A}$$

Knowing the current I_G , we can now work back and find I_{line} and I_{load} . Working back through T_1 , we get

$$\begin{aligned} N_{P1} I_G &= N_{S1} I_{\text{line}} \\ I_{\text{line}} &= \frac{N_{P1}}{N_{S1}} I_G \\ &= \frac{1}{10} (95.94 \angle -36.88^\circ \text{ A}) = 9.594 \angle -36.88^\circ \text{ A} \end{aligned}$$

**FIGURE 2-7**

(a) System with the load referred to the transmission system voltage level. (b) System with the load and transmission line referred to the generator's voltage level.

Working back through T_2 gives

$$\begin{aligned}
 N_{P2} \mathbf{I}_{\text{line}} &= N_{S2} \mathbf{I}_{\text{load}} \\
 \mathbf{I}_{\text{load}} &= \frac{N_{P2}}{N_{S2}} \mathbf{I}_{\text{line}} \\
 &= \frac{10}{1} (9.594 \angle -36.88^\circ \text{ A}) = 95.94 \angle -36.88^\circ \text{ A}
 \end{aligned}$$

It is now possible to answer the questions originally asked. The load voltage is given by

$$\begin{aligned}
 \mathbf{V}_{\text{load}} &= \mathbf{I}_{\text{load}} Z_{\text{load}} \\
 &= (95.94 \angle -36.88^\circ \text{ A})(5 \angle 36.87^\circ \Omega) \\
 &= 479.7 \angle -0.01^\circ \text{ V}
 \end{aligned}$$

and the line losses are given by

$$\begin{aligned}
 P_{\text{loss}} &= (I_{\text{line}})^2 R_{\text{line}} \\
 &= (9.594 \text{ A})^2 (0.18 \Omega) = 16.7 \text{ W}
 \end{aligned}$$

Notice that raising the transmission voltage of the power system reduced transmission losses by a factor of nearly 90! Also, the voltage at the load dropped much less in the system with transformers compared to the system without

transformers. This simple example dramatically illustrates the advantages of using higher-voltage transmission lines as well as the extreme importance of transformers in modern power systems.

Real power systems generate electric power at voltages in the range of 4 to 30 kV. They then use *step-up transformers* to raise the voltage to a much higher level (say 500 kV) for transmission over long distances, and *step-down transformers* to reduce the voltage to a reasonable level for distribution and final use. As we have seen in Example 2.1, this can greatly decrease transmission losses in the power system.